Chapter 15. Bott periodicity for the orthogonal group: complex structures and Clifford modules

Milnor, "Morse Theory", §24, rewritten for a first-year graduate student who wants to know what the theorem is for. Milnor calls his treatment "rather sketchy"; the missing details are supplied, and every proof is complete, except that, as in Chapter 14, coset fibrations, exact homotopy sequences and two facts about CW complexes are quoted.

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0. How many independent directions can be combed on a sphere?

On the sphere of directions S^2 no tangent vector field is nowhere zero (the hairy ball theorem, proved in Chapter 5 of the companion volume, *Topology from the Differentiable Viewpoint*, rewritten, Theorem 5.1). On the unit quaternions S^3 there are three at once, $x \mapsto ix, jx, kx$, orthonormal and tangent at every point; on S^7 octonion multiplication gives seven. How many pointwise independent tangent fields does S^{n-1} carry? The answer, due to Radon and Hurwitz (lower bound, 1922–23) and Adams (upper bound, 1962), is $\rho(n) - 1$, where for $n = 2^{4a+b} \cdot (\text{odd})$, $0 \le b \le 3$,

$$\rho(n) = 8a + 2^b.$$

The lower bound comes from linear algebra: if $J_1, \dots, J_k$ are orthogonal $n \times n$ matrices with $J_\alpha^2 = -I$ and $J_\alpha J_\beta = -J_\beta J_\alpha$, then $J_1 x, \dots, J_k x$ are orthonormal and perpendicular to x. The 8 in the formula is Bott's 8, $\pi_i O \cong \pi_{i+8} O$, and this is no coincidence: Milnor's proof of the periodicity theorem runs through these families of anti-commuting "complex structures". The minimal geodesics from I to $-I$ in $O(n)$ are the complex structures J_1; those from J_1 to $-J_1$ in the space of complex structures are the J_2 anti-commuting with J_1; and so on. After eight steps one is back at an orthogonal group, and Morse theory shifts homotopy groups by one at each step. §7 returns to vector fields and to real vector bundles over spheres.

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1. The orthogonal group and its minimal geodesics (Milnor pp. 133–136)

$O(n) = \{T \in M_n(\mathbb{R}) : TT^{\mathsf{T}} = I\}$. Chapter 14, §1, holds verbatim for real matrices: $O(n)$ is a compact Lie group with Lie algebra $\mathfrak{o}(n)$ of skew-symmetric matrices, the metric $\langle A, B\rangle = \operatorname{tr}(AB^{\mathsf{T}})$ induced from $M_n(\mathbb{R})$ (the restriction of that of $U(n)$) is bi-invariant, conjugations are isometries, and the geodesic through S with velocity SA is $t \mapsto S\exp(tA)$; geodesics from I stay in $SO(n)$. We quote the normal form (real spectral theorem, e.g. Axler): for $A \in \mathfrak{o}(2m)$ there is $T \in O(2m)$ with

$$TAT^{-1} = \operatorname{diag}(a_1 J_0, \dots, a_m J_0), \qquad J_0 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad a_j \ge 0,$$

and $\exp(\theta J_0) = \cos\theta\, I + \sin\theta\, J_0$ (Milnor p. 134).

Definition. For n even, a *complex structure* on $\mathbb{R}^n$ is a $J \in O(n)$ with $J^2 = -I$. $\Omega_1(n)$ denotes the set of complex structures.

Lemma 1.1. $J \in M_n(\mathbb{R})$ is a complex structure iff $J^{\mathsf{T}} = -J$ and $J^2 = -I$. For a complex structure, $\exp(\theta J) = \cos\theta\, I + \sin\theta\, J$, $|J|^2 = n$, and J is conjugate in $O(n)$ to $\operatorname{diag}(J_0, \dots, J_0)$.

Proof. If J is orthogonal, $J^{\mathsf{T}} = J^{-1} = -J$; if J is skew, $JJ^{\mathsf{T}} = -J^2 = I$. The series for exp uses $J^{2k} = (-1)^k I$, and $|J|^2 = \operatorname{tr}(JJ^{\mathsf{T}}) = n$. In the normal form, $TJ^2T^{-1} = \operatorname{diag}(-a_j^2 I_2) = -I$ forces every $a_j = 1$. □

Remark (Milnor p. 133). Fix $J_1 \in \Omega_1(n)$ and let $U(n/2) = \{T \in O(n) : TJ_1 = J_1T\}$, the unitary group of $\mathbb{R}^n$ made complex by $i = J_1$. Then $\Omega_1(n) \cong O(n)/U(n/2)$. Indeed $O(n)$ acts on $\Omega_1(n)$ by $T \cdot J = TJT^{-1}$, transitively by the last part of Lemma 1.1, with stabilizer $U(n/2)$ at J_1; apply (Q1) of Chapter 14.

Lemma 24.1. *(checked by machine: V15.26, restated)*[1] The space of minimal geodesics from I to $-I$ in $O(n)$ is homeomorphic to $\Omega_1(n)$, by $J \mapsto \gamma_J$, $\gamma_J(t) = \exp(\pi t J) = \cos(\pi t)\, I + \sin(\pi t)\, J$.

Proof. A geodesic from I to $-I$ is $\gamma(t) = \exp(\pi t A)$, $A \in \mathfrak{o}(n)$, $\exp(\pi A) = -I$; in the normal form this says every a_j is an odd integer. Its length is $|\pi A| = \pi(2\sum a_j^2)^{1/2} \geq \pi\sqrt{n}$, with equality iff every $a_j = 1$, i.e. $A^2 = -I$, i.e. $A \in \Omega_1(n)$. A minimal geodesic exists ($O(n)$ is compact; Milnor 10.9), so the minimal geodesics are the γ_J. The map $J \mapsto \gamma_J$ is injective ($J = \gamma_J'(0)/\pi$), continuous (constant speed, as in Chapter 14, Lemma 2.3), and $\Omega_1(n)$ is compact. □

Lemma 24.2. *(checked by machine: V15.3, restated; V15.30, in a narrower form)* Every non-minimal geodesic from I to $-I$ in $O(2m)$ has index $\geq 2m - 2$.

Proof. Conjugating by an orthogonal matrix (an isometry fixing $\pm I$, so preserving indices, as in Chapter 14, Lemma 4.1), take $A = \operatorname{diag}(a_1 J_0, \dots, a_m J_0)$, the a_j odd and positive, $a_1 \geq \cdots \geq a_m$. The velocity is $V = \pi A$, and by Theorems 21.3(b) and 20.5 the conjugate points are governed by $K_V = \pi^2 K_A$, $K_A(W) = -\frac{1}{4}[A,[A,W]]$.

Eigenvalues of K_A (Milnor's "computation shows"). Write $W \in \mathfrak{o}(2m)$ in 2×2 blocks, $W_{ji} = -W_{ij}^{\mathsf{T}}$, and split $M_2(\mathbb{R}) = C_+ \oplus C_-$, where $C_+ = \operatorname{span}(I, J_0)$ commutes with J_0 and $C_- = \operatorname{span}(\operatorname{diag}(1,-1), \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right))$ anti-commutes with it; $J_0 C_\pm = C_\pm$. For $i < j$ and $X \in C_\pm$ let $W_X^{(ij)}$ have block X at (i,j), $-X^{\mathsf{T}}$ at (j,i), zeros elsewhere. The (i,j) block of $[A,W]$ is $a_i J_0 W_{ij} - a_j W_{ij} J_0$, so $[A, W_X^{(ij)}] = W^{(ij)}_{(a_i \mp a_j) J_0 X}$ and $K_A W_X^{(ij)} = \frac{1}{4}(a_i \mp a_j)^2 W_X^{(ij)}$; the diagonal blocks lie in $\mathbb{R}J_0$ and commute with A. Since $4 \cdot m(m-1)/2 + m = \dim \mathfrak{o}(2m)$, the non-zero eigenvalues of K_A are $(a_i + a_j)^2/4$ ($i < j$) and $(a_i - a_j)^2/4$ ($a_i > a_j$), each twice.

Conjugate points. An eigenvalue e^2 of K_A gives conjugate points at $t = \pi s/\sqrt{\pi^2 e^2} = s/e$ (Theorem 20.5); for $e = (a_i \pm a_j)/2$, an integer, $e - 1$ of them lie in $(0,1)$. By the Index Theorem 15.1,

$$\lambda = \sum_{i<j} (a_i + a_j - 2) + \sum_{a_i > a_j} (a_i - a_j - 2),$$

all terms ≥ 0. For a minimal geodesic all $a_j = 1$ and $\lambda = 0$; otherwise $a_1 \geq 3$ and $\lambda \geq \sum_{j=2}^{m} (3 + 1 - 2) = 2m - 2$. □

[1] Checked by machine: the result has been verified in the Lean proof assistant. The numbers are those of the verified edition and of the comparison, which give the checked statement in full. *As stated*: the same statement. *Restated*: an equivalent statement in other terms. *With its hypotheses made explicit*: a hypothesis the book uses silently is stated. *In a stronger form*: the checked statement says more than the book's. *In a narrower form*: it covers less, a special setting or a weaker conclusion. *With a sign corrected*: the book's statement has a slip. *Not checked*: no part of the result has been verified. The section "Checked by machine" at the end of the chapter says which forms were checked, and covers the results that carry no such note.

Theorem 24.3 (Bott). *(checked by machine: V15.32, restated)* The inclusion $\Omega_1(n) \to \Omega(O(n); I, -I)$ of Lemma 24.1 induces isomorphisms of homotopy groups in dimensions $\leq n-4$. Hence $\pi_i\Omega_1(n) \cong \pi_{i+1}O(n)$ for $i \leq n-4$.

Proof. As in Chapter 14, Theorem 23.3, in the compact manifold $SO(n)$: $\Omega_1(n)$ is a manifold (the Remark, or Lemma 24.4), $\lambda_0 = n-2$ (Lemma 24.2), and $\pi_i\Omega(SO(n); I, -I) \cong \pi_{i+1}SO(n) = \pi_{i+1}O(n)$. □

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2. Anti-commuting complex structures and the spaces $\Omega_k(n)$ (Milnor pp. 136–138)

Definition. Fix complex structures $J_1, \dots, J_{k-1}$ on $\mathbb{R}^n$ that anti-commute, $J_rJ_s + J_sJ_r = 0$ for $r \neq s$ (they make $\mathbb{R}^n$ a module over a Clifford algebra, §5). Let $\Omega_k(n)$ be the set of complex structures anti-commuting with $J_1, \dots, J_{k-1}$, and $\Omega_0(n) = O(n)$: $\Omega_k(n) \subseteq \cdots \subseteq \Omega_1(n) \subseteq O(n)$, all compact.

Products of anti-commuting complex structures obey simple sign rules. Let $J_1, \dots, J_k$ anti-commute pairwise, and for $S = \{s_1 < \cdots < s_p\} \subseteq \{1, \dots, k\}$ put $J_S = J_{s_1} \cdots J_{s_p}$, $J_\emptyset = I$.

Lemma 2.1 (sign rules). (a) J_S is orthogonal and $J_S^{\mathsf{T}} = (-1)^{p(p+1)/2}J_S$: for $p \equiv 0, 3 \pmod 4$ it is symmetric with $J_S^2 = I$, for $p \equiv 1, 2 \pmod 4$ it is a complex structure. (b) $J_SJ_T = (-1)^{pq-|S\cap T|}J_TJ_S$ for $|T| = q$; in particular J_α commutes with J_S iff $p - [\alpha \in S]$ is even. (c) $J_SJ_T = \pm J_{S\triangle T}$, the sign depending only on S and T.

Proof. (a) $J_S^{\mathsf{T}} = (-1)^p J_{s_p} \cdots J_{s_1}$, and reversing p anti-commuting factors costs $(-1)^{p(p-1)/2}$; with $J_S^{\mathsf{T}}J_S = I$ this gives $J_S^2 = \pm I$ with the same sign. (b) Moving the q factors of J_T past the p factors of J_S costs a sign for each pair of distinct indices. (c) Bring equal factors together and use $J_\alpha^2 = -I$. □

Milnor calls a submanifold *totally geodesic* if each of its geodesics (for the induced metric) is a geodesic of the larger manifold.

Lemma 2.2. Let N be a submanifold of a Riemannian manifold M, with the induced metric. (a) If for every $v \in TN_x$ the M-geodesic $t \mapsto \exp_x^M(tv)$ stays in N for small $|t|$, then N is totally geodesic. (b) If N is totally geodesic, then $R^N(X,Y)Z = R^M(X,Y)Z$ for $X, Y, Z \in TN_x$. (c) An isometry of M that maps N onto N restricts to an isometry of N.

Proof. (a) Let $c(t) = \exp_x^M(tv)$ lie in N for $|t| < \varepsilon$. For small δ, $c|[0,\delta]$ is no longer than any path in M between its endpoints (Theorem 10.4), in particular than any path in N, which has the same length in both. So it is a minimal path of N parametrized proportionally to arc length, hence a geodesic of N (Corollary 10.7, applied in N). By uniqueness, the N-geodesic with velocity v coincides with c near $t = 0$; as x and v are arbitrary, every N-geodesic is locally, hence globally, an M-geodesic. (b) For X, Y tangent to N, the tangential part of $\nabla_X^M Y$ is a symmetric connection compatible with the metric, hence is $\nabla_X^N Y$ (Lemma 8.6). The rest, $\mathrm{II}(X,Y) = \nabla_X^M Y - \nabla_X^N Y$, is tensorial and symmetric, and the same splitting holds along curves. Along an N-geodesic c, an M-geodesic by hypothesis, $0 = D^M c'/dt = \mathrm{II}(c', c')$; so $\mathrm{II} = 0$ by polarization, $\nabla^M = \nabla^N$ on fields tangent to N, and $R(X,Y)Z = -\nabla_X\nabla_Y Z + \nabla_Y\nabla_X Z + \nabla_{[X,Y]}Z$ is the same in N and M. (c) Its inverse also preserves N. □

Lemma 24.4. *(checked by machine: V15.23, restated)* (i) Each $\Omega_k(n)$ is a smooth, totally geodesic

submanifold of $O(n)$; its tangent space at J is $J\mathcal{T}_J$, where

$$\mathcal{T}_J = \{A \in \mathfrak{o}(n) : AJ = -JA,\ AJ_\alpha = J_\alpha A \text{ for } \alpha < k\}.$$

(ii) Each component of $\Omega_k(n)$ is a symmetric space. (iii) Let $J_k \in \Omega_k(n)$ (and $J_0 = I$), let $\Omega_{k+1}(n)$ be the set of $J' \in \Omega_k(n)$ anti-commuting with J_k, and suppose $\Omega_{k+1}(n) \neq \emptyset$. Then the minimal geodesics from J_k to $-J_k$ in $\Omega_k(n)$ are exactly the semicircles

$$\gamma_{J'}(t) = \cos(\pi t)\, J_k + \sin(\pi t)\, J', \qquad J' \in \Omega_{k+1}(n),$$

and $J' \mapsto \gamma_{J'}$ is a homeomorphism of $\Omega_{k+1}(n)$ onto the space of minimal geodesics.

For $k = 0$, (i) is Chapter 14's chart argument and (iii) is Lemma 24.1; assume $k \geq 1$. Milnor's version of (iii), "for $0 \leq l < k$", tacitly needs $\Omega_{l+1}(n) \neq \emptyset$, which his standing assumption provides.

Proof. Let $B_\varepsilon \subseteq \mathfrak{o}(n)$ be a ball on which exp is a diffeomorphism onto a neighbourhood of I in $O(n)$, invariant under $A \mapsto -A$ and orthogonal conjugation. Let $J \in \Omega_k(n)$ and $A \in \mathfrak{o}(n)$.

Assertion 1. If $AJ = -JA$, then $J \exp A$ is a complex structure; for $A \in B_\varepsilon$ the converse holds. Indeed $(J \exp A)^2 = J^2(J^{-1}\exp(A)J)\exp A = -\exp(J^{-1}AJ)\exp(A)$, which is $-I$ iff $\exp(J^{-1}AJ) = \exp(-A)$; this holds if $J^{-1}AJ = -A$, and for $A \in B_\varepsilon$ only then.

Assertion 2. If $AJ_\alpha = J_\alpha A$, then $J \exp A$ anti-commutes with J_α; for $A \in B_\varepsilon$ the converse holds. Indeed $J_\alpha J \exp A = -J \exp(J_\alpha A J_\alpha^{-1}) J_\alpha$, which is $-(J \exp A)J_\alpha$ iff $\exp(J_\alpha A J_\alpha^{-1}) = \exp A$.

(i) So $\Omega_k(n) \cap J\exp(B_\varepsilon) = J\exp(B_\varepsilon \cap \mathcal{T}_J)$; as $A \mapsto J\exp A$ is a chart of $O(n)$, $\Omega_k(n)$ is near J an open piece of the linear subspace $\mathcal{T}_J$ in this chart (components may have different dimensions). The $O(n)$-geodesic $t \mapsto J\exp(tA)$, $A \in \mathcal{T}_J$, stays in $\Omega_k(n)$ for all t by the "if" halves, so $\Omega_k(n)$ is totally geodesic by Lemma 2.2(a): this is Milnor's "clearly implies".

(ii) The reflection of $O(n)$ in J is $I_J(S) = JS^{-1}J$ (Lemma 21.1), and for $J' \in \Omega_k(n)$, $I_J(J') = -JJ'J$ is orthogonal, squares to $JJ'J^2J'J = J^2 = -I$ and anti-commutes with each J_α. So I_J is an isometry of $\Omega_k(n)$ (Lemma 2.2(c)) fixing J with derivative -1, reversing the geodesics through J; $\Omega_k(n)$ is compact, so each component is a symmetric space.

(iii) Distances in $\Omega_k(n)$ are $\geq$ distances in $O(n)$, and $\rho_O(J_k, -J_k) = \rho_O(I, -I) = \pi\sqrt{n}$. For $J' \in \Omega_{k+1}(n)$ let $A = J_k^{-1}J' = -J_kJ'$: then $A^2 = -J_k^2J'^2 = -I$, A anti-commutes with J_k and commutes with each J_α, $\alpha < k$, so $A \in \mathcal{T}_{J_k}$ is a complex structure and $J_k\exp(\pi tA) = \gamma_{J'}(t)$ (Lemma 1.1). This $O(n)$-geodesic lies in $\Omega_k(n)$ (Assertions 1, 2) and has length $\pi\sqrt{n}$, so it is a minimal path, hence a minimal geodesic, of $\Omega_k(n)$ (Corollary 10.7), and the distance in $\Omega_k(n)$ is $\pi\sqrt{n}$. Conversely, a minimal geodesic γ of $\Omega_k(n)$ from J_k to $-J_k$ is an $O(n)$-geodesic of length $\pi\sqrt{n}$, so $J_k^{-1}\gamma$ is minimal from I to $-I$: $\gamma(t) = J_k\exp(\pi tA)$ with A a complex structure (Lemma 24.1). As $\gamma(t) \in \Omega_k(n)$ for small t, the "only if" halves give $AJ_k = -J_kA$, $AJ_\alpha = J_\alpha A$, and then $J' = J_kA$ is in $\Omega_{k+1}(n)$ ($J'^2 = -J_k^2A^2 = -I$, $J_kJ' = -A = -J'J_k$, and J' anti-commutes with J_α because J_k does and A commutes with it), with $\gamma = \gamma_{J'}$. The homeomorphism follows as in Lemma 24.1. □

Remark (Milnor p. 138). The point J' corresponding to a minimal geodesic is its midpoint $\gamma_{J'}(1/2)$. All of $\Omega_k(n)$ lies on the sphere $|J|^2 = n$ in $M_n(\mathbb{R})$, and $\langle J', J_k\rangle = -\operatorname{tr}(J'J_k) = 0$ since $J'J_k$ is skew: the minimal geodesics are great semicircles from the "pole" J_k to $-J_k$ through the "equator" $\Omega_{k+1}(n)$, as in the picture of §22 for antipodal points of S^{n+1}.

3. Stabilization and the statement of the periodicity theorem (Milnor pp. 138, 142)

To pass to the limit we need structures on a block that is added again and again. Let $\sigma = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$, $\tau = \mathrm{diag}(1,-1)$, $\kappa = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$; any two anti-commute, $\sigma^2 = -I$, $\tau^2 = \kappa^2 = I$. On $\mathbb{R}^{16} = (\mathbb{R}^2)^{\otimes 4}$ (Kronecker products) put

$$J'_1 = 1\otimes 1\otimes \sigma\otimes\tau, \quad J'_2 = 1\otimes\sigma\otimes\tau\otimes\tau, \quad J'_3 = \sigma\otimes 1\otimes\kappa\otimes\tau, \quad J'_4 = \sigma\otimes\tau\otimes\tau\otimes\tau,$$
$$J'_5 = \sigma\otimes\kappa\otimes\tau\otimes\tau, \quad J'_6 = \tau\otimes\sigma\otimes\kappa\otimes\tau, \quad J'_7 = \kappa\otimes\sigma\otimes\kappa\otimes\tau, \quad J'_8 = 1\otimes 1\otimes 1\otimes\sigma.$$

Products are computed slot by slot, so two such words anti-commute iff they carry different non-identity letters in an odd number of slots, and a word with one σ is skew with square $-I$; checking the 28 pairs shows that $J'_1, \dots, J'_8$ are anti-commuting complex structures.

For $n = 16r$ let $J_\alpha = J'_\alpha \oplus \cdots \oplus J'_\alpha$ (r copies) on $\mathbb{R}^{16r} = (\mathbb{R}^{16})^r$. From now on $\Omega_k(16r)$ is defined with these $J_1, \dots, J_{k-1}$, with base point J_k. The map $J \mapsto J \oplus J'_k$ (for $k = 0$, $T \mapsto T \oplus I_{16}$) embeds $\Omega_k(16r)$ in $\Omega_k(16r+16)$, carrying J_k to J_k.

Definition (Milnor p. 138). $\Omega_k = \bigcup_r \Omega_k(16r)$ with the colimit topology. The space $O = \Omega_0$ is the *infinite orthogonal group*. (Milnor uses all n; the multiples of 16 are cofinal.)

The maps $\varphi_r : \Omega_{k+1}(16r) \to \Omega(\Omega_k(16r); J_k, -J_k)$, $\varphi_r(J') = \gamma_{J'}$, of Lemma 24.4(iii) are compatible with the embeddings:

$$\varphi_{r+1}(J' \oplus J'_{k+1}) = \varphi_r(J') \oplus \delta_k, \qquad \delta_k(t) = \cos(\pi t)\, J'_k + \sin(\pi t)\, J'_{k+1},$$

where on path spaces the embedding is $\omega \mapsto \omega \oplus \delta_k$. This is what Milnor calls "not difficult to see": the inclusions pass to a limit map $\Omega_{k+1} \to \Omega\Omega_k$. By the lemma on compact sets in a colimit (Chapter 14, proof of Corollary 23.6), $\pi_i\Omega_k = \mathrm{colim}_r\, \pi_i\Omega_k(16r)$, and similarly for the spaces of continuous paths from J_k to $-J_k$, which we denote $\Omega\Omega_k$.

Theorem 24.5. *(checked by machine: V15.43, restated)* For $0 \le k \le 7$ the limit map $\varphi : \Omega_{k+1} \to \Omega\Omega_k$ induces a bijection of path components and isomorphisms of all homotopy groups. Consequently, with base points J_k,

$$\pi_h O \cong \pi_{h-1}\Omega_1 \cong \pi_{h-2}\Omega_2 \cong \cdots \cong \pi_1\Omega_{h-1} \cong \pi_0\Omega_h \qquad (1 \le h \le 8),$$

the last bijection being one of sets.

The consequence holds because the paths from J_k to $-J_k$ form a space homotopy equivalent to the loops at J_k, compatibly (up to homotopy) with the embeddings, so $\pi_j\Omega\Omega_k \cong \pi_{j+1}\Omega_k$ (Chapter 14, proof of Theorem 23.3). Milnor states Theorem 24.5 for all $k \ge 0$, but his proof needs Lemma 24.8 for $k+1$ structures, which he gives only for $k+1 \le 8$; $k \le 7$ is all that periodicity needs. Milnor's conclusion is a homotopy equivalence, which needs the two facts quoted at the end of §6; his chain of isomorphisms stops at $\pi_1\Omega_{h-1}$, and the last bijection is added here. The proof is in §6.

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4. The eight spaces (Milnor pp. 138–142)

Given $J_1, \dots, J_7$, define

$$\theta_3 = J_1J_2J_3, \qquad \theta_5 = J_1J_4J_5, \qquad \theta_6 = J_2J_4J_6, \qquad \theta_7 = J_1J_6J_7.$$

By Lemma 2.1 these are symmetric orthogonal involutions; any two commute (their index sets meet in one element), and J_α commutes with θ_{abc} iff $\alpha \in \{a,b,c\}$. Solving for the last factor, $J_3 = -J_1J_2\theta_3$, $J_5 = J_4J_1\theta_5$, $J_6 = J_4J_2\theta_6$, $J_7 = J_6J_1\theta_7$. The table records which J_α commute (c) or anti-commute (a) with which θ:

	θ_3	θ_5	θ_6	θ_7
J_1	c	c	a	c
J_2	c	a	c	a
J_3	c	a	a	a
J_4	a	c	c	a
J_5	a	c	a	a
J_6	a	a	c	c
J_7	a	a	a	c
J_8	a	a	a	a

Lemma 24.6 rests on two observations.

Lemma 4.1 (odd steps). Let $J_1, \dots, J_{k-1}$ be fixed and $a < b < k$. Then $J \mapsto E = \{v : J_aJ_bJv = v\}$ is a bijection from $\Omega_k(n)$ onto the set of subspaces E with $J_aE = E$, $J_bE = E$ and $J_cE = E^\perp$ for all $c < k$, $c \notin \{a,b\}$. The inverse is $J = J_bJ_a(2P_E - I)$, P_E the orthogonal projection, so the bijection is a homeomorphism when subspaces are topologized by their projections.

Proof. For $J \in \Omega_k(n)$, $\theta = J_aJ_bJ$ is a symmetric involution commuting with J_a, J_b and anti-commuting with the other J_c (Lemma 2.1), so $\theta = 2P_E - I$ with E as stated. Conversely, $\theta = 2P_E - I$ commutes with J_a, J_b and, since $J_cE = E^\perp$ forces $J_cE^\perp = E$, anti-commutes with J_c; then $J = J_bJ_a\theta$ is orthogonal, $J^2 = (J_bJ_a)^2\theta^2 = -I$, and J anti-commutes with J_a, J_b (as J_bJ_a does) and with J_c (as θ does). Finally $\theta = J_aJ_bJ$ iff $J = J_bJ_a\theta$. □

Lemma 4.2 (even steps). For $k \geq 2$, $J \mapsto Q = J_{k-1}J$ is a bijection from $\Omega_k(n)$ onto the set of $Q \in O(n)$ with $Q^2 = -I$ that commute with $J_1, \dots, J_{k-2}$ and anti-commute with J_{k-1}; the inverse is $Q \mapsto -J_{k-1}Q$.

Proof. Lemma 2.1 gives the properties of Q. Conversely, for $J = -J_{k-1}Q$: $J^2 = J_{k-1}QJ_{k-1}Q = -J_{k-1}^2Q^2 = -I$; for $c \leq k-2$, $J_cJ = J_{k-1}J_cQ = J_{k-1}QJ_c = -JJ_c$; and $J_{k-1}J = Q = -JJ_{k-1}$. □

The bijections of Lemmas 4.1 and 4.2, and those built from them in Lemma 24.6 below, are restrictions of affine maps between spaces of matrices (subspaces enter through their projections, and an isometry T has $T^{-1} = T^\mathsf{T}$), with inverses of the same kind; so they are diffeomorphisms between the submanifolds involved, as Milnor states.

Ω_2 (Milnor pp. 138–139). With $i = J_1$, $\mathbb{R}^n = \mathbb{C}^{n/2}$, and $J \in \Omega_2(n)$ is a conjugate-linear complex structure; J_1, J, J_1J satisfy the relations of the quaternion units, a "quaternionic structure". For $4 \mid n$, $\Omega_2(n) \cong U(n/2)/Sp(n/4)$, with $Sp(n/4)$ the isometries commuting with J_1, J_2: $U(n/2)$ acts by conjugation, transitively by Lemma 24.8 below ($k = 2$), with stabilizer $Sp(n/4)$ at J_2.

From now on $n = 16r$, with the structures of §3. In $\mathbb{R}^{16}$ the sixteen joint eigenspaces of $\theta_3, \theta_5, \theta_6, \theta_7$ are lines (this follows from §5: $\mathbb{R}^{16}$ is a minimal $(J_1', \dots, J_8')$-space, and the proof of Lemma 24.8 shows that all sixteen are non-zero), so in $\mathbb{R}^{16r}$ they all have dimension r.

Lemma 4.3 (the pieces). (a) Once $J_1, \dots, J_4$ are fixed, $\mathbb{R}^{16r} = V_1 \oplus J_4 V_1$, $V_1 = \{\theta_3 = 1\}$. (b) Once $J_1, \dots, J_5$ are fixed, with $W = V_1 \cap \{\theta_5 = 1\}$, $\mathbb{R}^{16r} = W \oplus J_2 W \oplus J_4 W \oplus J_4 J_2 W$ and $V_1 = W \oplus J_2 W$. (c) Once $J_1, \dots, J_6$ are fixed, with $X = W \cap \{\theta_6 = 1\}$, $\mathbb{R}^{16r} = \bigoplus_{S \subseteq \{1,2,4\}} J_S X$ and $W = X \oplus J_1 X$. All sums are orthogonal, the summands being the joint eigenspaces of the θ's involved.

Proof. Commuting symmetric involutions split $\mathbb{R}^{16r}$ orthogonally into joint eigenspaces; by the table, among $\theta_3, \theta_5, \theta_6$, J_4 reverses the sign of θ_3 only, J_2 of θ_5 only, J_1 of θ_6 only, and so permutes the joint eigenspaces accordingly. □

Lemma 24.6-(3). *(checked by machine: V15.11, in a narrower form)* $J_3 \mapsto V_1 = \{\theta_3 = 1\}$ identifies $\Omega_3(16r)$ with the set of all quaternionic subspaces of $\mathbb{H}^{4r}$ (subspaces invariant under J_1 and J_2). On V_1, $J_3 = -J_1 J_2$; on $V_2 = V_1^\perp$, $J_3 = J_1 J_2$.

Proof. Lemma 4.1 with $(a, b) = (1, 2)$ and no further c; $J_3 = J_2 J_1 (2P - I)$. □

$\Omega_3(16r)$ has components $G_j(\mathbb{H}^{4r})$ of varying dimension $4j(4r - j)$. Milnor restricts to the largest, $j = 2r$: $\dim_{\mathbb{H}} V_1 = \dim_{\mathbb{H}} V_2 = 2r$, as for our J_3.

Lemma 24.6-(4). $J_4 \mapsto T = J_3 J_4 | V_1$ identifies $\Omega_4(16r)$ with the set of quaternionic isometries $T : V_1 \to V_2$, the inverse being $J_4|V_1 = J_3^{-1} T$, $J_4|V_2 = -T^{-1} J_3$. Thus $\Omega_4(16r) \cong Sp(2r)$.

Proof. By Lemma 4.2, $J_4 \leftrightarrow Q = J_3 J_4$ with $Q^2 = -I$, Q commuting with J_1, J_2 and anti-commuting with J_3, hence with θ_3. So Q maps V_1 to V_2 and back, $T = Q|V_1$ is an isometry commuting with J_1, J_2, and $Q|V_2 = -T^{-1}$. Conversely, for such a T, $Q = T \oplus (-T^{-1})$ satisfies $Q^2 = -I$, commutes with J_1, J_2 (so with $J_1 J_2$), and anti-commutes with θ_3, hence with $J_3 = -J_1 J_2 \theta_3$. Then $J_4 = -J_3 Q$, which is $J_3^{-1} T$ on V_1 and $J_3 T^{-1} = -T^{-1} J_3$ on V_2 (T intertwines $J_3|V_1 = -J_1 J_2$ with $-J_3|V_2$). Choosing a base point T_0, $T \mapsto T_0^{-1} T$ identifies the set with the group of quaternionic isometries of $V_1 \cong \mathbb{H}^{2r}$. □

Lemma 24.6-(5). *(checked by machine: V15.11, in a narrower form)* $J_5 \mapsto W = V_1 \cap \{\theta_5 = 1\}$ identifies $\Omega_5(16r)$ with the set of subspaces $W \subseteq V_1$ such that (1) $J_1 W = W$ and (2) $V_1 = W \oplus J_2 W$, orthogonally.

Proof. Lemma 4.1 with $(a, b) = (1, 4)$: $\Omega_5 \leftrightarrow E$ with $J_1 E = J_4 E = E$, $J_2 E = J_3 E = E^\perp$. Given E: θ_5 commutes with θ_3, so $E = (E \cap V_1) \oplus (E \cap V_2)$ with $E \cap V_2 = J_4 W$; $J_1 W = W$; and J_2, preserving V_1, maps W onto $E^\perp \cap V_1$. Conversely, given W, put $E = W \oplus J_4 W$: then $J_1 E = J_4 E = E$, $J_2 E = J_2 W \oplus J_4 J_2 W = E^\perp$ by Lemma 4.3(a), and $J_3 E = J_1 J_2 \theta_3 E = E^\perp$. □

Remark. $\Omega_5(16r) \cong Sp(2r)/U(2r)$, where $U(2r)$ is the group of quaternionic automorphisms of V_1 preserving a fixed W. Indeed $Sp(V_1)$ acts transitively on the set of such W: a unitary isomorphism $U : W \to W'$ (for $i = J_1$) extends to V_1 by $J_2 w \mapsto J_2 U w$, and this extension commutes with J_1 and J_2; the stabilizer of W restricts isomorphically to the unitary group of W.

Lemma 24.6-(6). *(checked by machine: V15.11, in a narrower form)* $J_6 \mapsto X = W \cap \{\theta_6 = 1\}$ identifies $\Omega_6(16r)$ with the set of real subspaces $X \subseteq W$ such that $W = X \oplus J_1 X$, orthogonally.

Proof. Lemma 4.1 with $(a, b) = (2, 4)$: $\Omega_6 \leftrightarrow E$ with $J_2 E = J_4 E = E$, $J_1 E = J_3 E = J_5 E = E^\perp$. Given E: θ_6 commutes with θ_3, θ_5, so E splits along the four pieces of Lemma 4.3(b), and as J_2, J_4 preserve E, $E = X \oplus J_2 X \oplus J_4 X \oplus J_4 J_2 X$; J_1 maps X onto $E^\perp \cap W$. Conversely, given X, define E by this formula: J_2, J_4 permute its summands; $J_1 E$ is the same sum built on $J_1 X = W \ominus X$,

which is $E^\perp$; and since θ_3, θ_5 are ± 1 on each piece, $J_3 E = J_1 J_2 E = E^\perp$, $J_5 E = J_4 J_1 E = E^\perp$. $\square$

Remark. $\Omega_6(16r) \cong U(2r)/O(2r)$: the unitary group of (W, J_1) acts transitively on such X (a real isometry $X \to X'$ extends complex-linearly), with stabilizer $O(X)$.

Lemma 24.6-(7). *(checked by machine: V15.11, in a narrower form)* $J_7 \mapsto X_1 = X \cap \{\theta_7 = 1\}$ identifies $\Omega_7(16r)$ with the set of all subspaces of $X \cong \mathbb{R}^{2r}$.

Proof. Lemma 4.1 with $(a, b) = (1, 6)$: $\Omega_7 \leftrightarrow E$ with $J_1 E = J_6 E = E$ and $J_c E = E^\perp$ for $c = 2, \dots, 5$. Given E: θ_7 commutes with $\theta_3, \theta_5, \theta_6$, so E splits along the eight pieces $J_S X$; as J_1 preserves E while J_2, J_4 exchange E and $E^\perp$,

$$E = \bigoplus_{S \subseteq \{1,2,4\}} J_S X_{(S)}, \qquad X_{(S)} = X_1 \text{ if } |S \cap \{2,4\}| \text{ is even}, \quad X_{(S)} = X_2 = X \ominus X_1 \text{ otherwise.} \tag{4.1}$$

Conversely, for any $X_1 \subseteq X$ define E by (4.1): $J_1 E = E$, $J_2 E = J_4 E = E^\perp$, $J_6 E = J_4 J_2 \theta_6 E = E$, $J_3 E = J_1 J_2 \theta_3 E = E^\perp$, $J_5 E = J_4 J_1 \theta_5 E = E^\perp$. $\square$

Again the components $G_j(\mathbb{R}^{2r})$ vary in dimension, and Milnor restricts to $\dim X_1 = \dim X_2 = r$, as for our J_7. **Assertion (Milnor p. 141):** the largest component of $\Omega_7(16r)$ is diffeomorphic to the Grassmann manifold of r-dimensional subspaces of $\mathbb{R}^{2r}$.

Lemma 24.6-(8). *(checked by machine: V15.42, restated)* $J_8 \mapsto T = J_7 J_8 | X_1$ identifies $\Omega_8(16r)$ with the set of isometries $T : X_1 \to X_2$. Thus $\Omega_8(16r) \cong O(r)$.

Proof. By Lemma 4.2, $J_8 \leftrightarrow Q = J_7 J_8$, $Q^2 = -I$, commuting with $J_1, \dots, J_6$ and anti-commuting with J_7. Then Q commutes with $\theta_3, \theta_5, \theta_6$ and with J_1, J_2, J_4, so it preserves X and is determined by $Q_X = Q|X$ through $Q(J_S x) = J_S Q_X x$; and it anti-commutes with θ_7, so Q_X exchanges X_1 and X_2, with $Q_X | X_2 = -T^{-1}$. Conversely, given T, put $Q_X = T \oplus (-T^{-1})$ and $Q = J_S Q_X J_S^{-1}$ on $J_S X$. Then $Q^2 = -I$; Q commutes with J_1, J_2, J_4 (as $J_\alpha J_S = \pm J_{S \triangle \{\alpha\}}$) and with $\theta_3, \theta_5, \theta_6$ (scalars on each piece), hence with J_3, J_5, J_6; and it anti-commutes with θ_7 (on $J_S X$, $\theta_7 = (-1)^{|S \cap \{2,4\}|} J_S \theta_7 J_S^{-1}$, and Q_X anti-commutes with $\theta_7 | X$), hence with $J_7 = J_6 J_1 \theta_7$. $\square$

Passing to the limit. Under $J \mapsto J \oplus J_8'$ the spaces X_1, X_2 of $\mathbb{R}^{16r+16}$ are $X_1 \oplus X_1'$, $X_2 \oplus X_2'$ (lines $X_1', X_2' \subseteq \mathbb{R}^{16}$), and $T \mapsto T \oplus T_0'$ with $T_0' = J_7' J_8' | X_1'$. With the base point $T_0 = J_7 J_8 | X_1$ and $T \mapsto T_0^{-1} T$, this is $O(r) \to O(r+1)$, $T \mapsto T \oplus 1$. Hence Ω_8 is homeomorphic to O, compatibly with base points. Likewise $\Omega_4(16r) \cong Sp(2r)$ is compatible with $Sp(2r) \to Sp(2r+2)$, and $Sp = \Omega_4$ is the infinite symplectic group.

Remark (Milnor's footnote, p. 142). For $k > 8$, with $J_9, \dots, J_{k-1}$ also fixed, $J \mapsto J_8 J | X_1$ identifies $\Omega_k(16r)$ with $\Omega_{k-8}(r)$, defined on $X_1 \cong \mathbb{R}^r$ with the fixed structures $J_8 J_\alpha | X_1$, $9 \le \alpha < k$. Indeed $J_8 J$ commutes with $J_1, \dots, J_7$, hence preserves X_1, and is a complex structure anti-commuting with each $J_8 J_\alpha$ (Lemma 2.1). Conversely, $\mathbb{R}^{16r} = \bigoplus_{S \subseteq \{1,2,4,8\}} J_S X_1$ (the flips of J_1, J_2, J_4, J_8 on $\theta_3, \theta_5, \theta_6, \theta_7$ form a basis of $(\mathbb{Z}/2)^4$), and an operator C on X_1 extends uniquely to $\hat{C}$ commuting with $J_1, \dots, J_7$ and anti-commuting with J_8: $\hat{C}(J_S x) = \pm J_S C x$, with $-$ iff $8 \in S$; then $J = -J_8 \hat{C}$. The same decomposition shows that a space V with $k+8$ anti-commuting structures has $\dim V = 16 \dim X_1(V)$, with V minimal iff $X_1(V)$ is; and k structures C_β on X give $k+8$ on $\mathbb{R}^{16} \otimes X$, namely $J_\alpha' \otimes 1$ and $\omega_8' \otimes C_\beta$, where $\omega_8' = J_1' \cdots J_8'$ is a symmetric involution anti-commuting with each J_α'. Hence $m_{k+8} = 16 m_k$.

Theorem 24.7 (Bott). *(checked by machine: V15.45, in a narrower form)* $\pi_i O \cong \pi_{i+8} O$ for $i \geq 0$. With the facts quoted at the end of §6, O has the homotopy type of its eighth loop space $\Omega^8 O$, $O \simeq \Omega^4 Sp$ and $Sp \simeq \Omega^4 O$.

Proof. Theorem 24.5, applied eight times, and $\Omega_8 \cong O$: $\pi_{i+8} O \cong \pi_{i+7}\Omega_1 \cong \cdots \cong \pi_i \Omega_8 \cong \pi_i O$. Similarly $\Omega_4 \simeq \Omega^4 \Omega_0$ and $\Omega_8 \simeq \Omega^4 \Omega_4$. □

The table (Milnor p. 142, "left to the reader"). By Theorem 24.5, $\pi_h O \cong \pi_0 \Omega_h$ as sets for $1 \leq h \leq 8$, and $\pi_0 \Omega_h$ is visible from Lemma 24.6, using the connectedness of $U(n)$, $Sp(n)$, $SO(n)$ and of the Grassmannians:

k	$\Omega_k(16r)$	limit	$\pi_0 \Omega_k$
0	$O(16r)$	O	$\mathbb{Z}/2$
1	$O(16r)/U(8r)$	O/U	$\mathbb{Z}/2$
2	$U(8r)/Sp(4r)$	U/Sp	0
3	$\coprod_j G_j(\mathbb{H}^{4r})$	$\mathbb{Z} \times BSp$	$\mathbb{Z}$
4	$Sp(2r)$	Sp	0
5	$Sp(2r)/U(2r)$	Sp/U	0
6	$U(2r)/O(2r)$	U/O	0
7	$\coprod_j G_j(\mathbb{R}^{2r})$	$\mathbb{Z} \times BO$	$\mathbb{Z}$
8	$O(r)$	O	$\mathbb{Z}/2$

($O(n)/U(n/2)$ has two components because $U(n/2) \subseteq SO(n)$ is connected; in the limit the components of Ω_3 are labelled by $j - 2r \in \mathbb{Z}$, since $J \mapsto J \oplus J_3'$ adds 2 to j, and those of Ω_7 by $j - r$; BSp and BO name the colimits of the middle Grassmannians.) So $\pi_i O$ for $i \equiv 0, \ldots, 7 \pmod 8$ is $\mathbb{Z}/2, \mathbb{Z}/2, 0, \mathbb{Z}, 0, 0, 0, \mathbb{Z}$, and $\pi_i Sp = \pi_i \Omega_4$ is $0, 0, 0, \mathbb{Z}, \mathbb{Z}/2, \mathbb{Z}/2, 0, \mathbb{Z}$, Milnor's table. For the group structures: a group of order two is $\mathbb{Z}/2$; the bundles $Sp(m-1) \to Sp(m) \to S^{4m-1}$ make $\pi_j Sp(m-1) \to \pi_j Sp(m)$ an isomorphism for $j \leq 4m-3$, so $\pi_j Sp \cong \pi_j Sp(1) = \pi_j S^3$ for $j \leq 5$ and $\pi_7 O \cong \pi_3 \Omega_4 = \pi_3 Sp \cong \mathbb{Z}$; and $\pi_3 O \cong \pi_1 \Omega_2 = \pi_1(U/Sp) \cong \pi_1 U \cong \mathbb{Z}$, from the bundle $Sp(4r) \to U(8r) \to U(8r)/Sp(4r)$ and $\pi_0 Sp = \pi_1 Sp = 0$. Milnor's hint is a check: $Sp(1) = S^3$, and $SO(3) \cong \mathbb{R}P^3$ gives $\pi_1 O = \pi_1 SO(3) = \mathbb{Z}/2$.

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5. Clifford modules (Milnor pp. 142–144)

Let V be a Euclidean space with anti-commuting complex structures $J_1, \ldots, J_k$ (a $(J_1, \ldots, J_k)$-space). **Definition (Milnor p. 143).** V is *minimal* if no proper non-trivial subspace is closed under $J_1, \ldots, J_k$. Two minimal spaces are *isomorphic* if there is an isometry between them commuting with the J_α.

Lemma 5.1. Every $(J_1, \ldots, J_k)$-space is an orthogonal sum of minimal ones.

Proof. The J_α are skew, so the orthogonal complement of an invariant subspace is invariant. Induct on the dimension. □

Lemma 5.2 (cyclic vectors). Let V be minimal and $u \in V$ a unit vector. Then the $J_S u$, $S \subseteq \{1, \ldots, k\}$, span V. If V' is minimal, with a unit vector u' such that $\langle J_S u, u \rangle = \langle J_S u', u' \rangle$ for all S, then $\sum c_S J_S u \mapsto \sum c_S J_S u'$ is a well-defined isomorphism $V \to V'$.

Proof. The span is invariant (Lemma 2.1(c)), hence all of V. By Lemma 2.1(a),(c), $\langle J_S u, J_T u\rangle = \langle J_T^{\mathsf{T}} J_S u, u\rangle = \pm\langle J_{S\triangle T} u, u\rangle$ with a sign depending only on S, T. So the two families have the same Gram matrices, the map is well defined and isometric, and it commutes with J_α because $J_\alpha J_S = \pm J_{S\triangle\{\alpha\}}$ with the same sign in V and V'. □

Lemma 24.8 (Bott and Shapiro), for $k \le 8$. *(checked by machine: V15.17, in a narrower form)* For $0 \le k \le 8$, minimal $(J_1, \dots, J_k)$-spaces exist and have dimension m_k:

$$m_0 = 1,\ m_1 = 2,\ m_2 = m_3 = 4,\ m_4 = m_5 = m_6 = m_7 = 8,\ m_8 = 16.$$

For $k \not\equiv 3 \pmod 4$ any two are isomorphic. For $k = 3, 7$ there are exactly two isomorphism classes, distinguished by $\omega_k = J_1 \cdots J_k = \pm I$.

Proof. Let Θ_k consist of those of $\theta_3, \theta_5, \theta_6, \theta_7$ whose indices are all $\le k$, and let $t_k = |\Theta_k|$ ($0, 0, 0, 1, 1, 2, 3, 4, 4$ for $k = 0, \dots, 8$). Write G_k for the index sets S with $J_S = \pm$ a product of elements of Θ_k; the generators $\{1,2,3\}$, $\{1,4,5\}$, $\{2,4,6\}$, $\{1,6,7\}$ are independent under $\triangle$, so $|G_k| = 2^{t_k}$.

Step 1: the commutant. If $S \subseteq \{1, \dots, k\}$, J_S is symmetric and commutes with all of Θ_k, then $S \in G_k$. By Lemma 2.1(b), J_S commutes with θ_T ($|T| = 3$) iff $|S \setminus T|$ is even, a linear condition over $\mathbb{Z}/2$ on the indicator vector of S; and J_S is symmetric iff $|S| \equiv 0, 3 \pmod 4$. For $k \le 2$ only $S = \emptyset$ is symmetric. For $k = 3$ the symmetric sets are $\emptyset$ and $\{1,2,3\}$. For $k = 4$ the condition is $4 \notin S$, leaving $\emptyset$ and $\{1,2,3\}$. For $k = 5$ the conditions are that $|S \cap \{4,5\}|$ and $|S \cap \{2,3\}|$ be even; of the eight solutions the symmetric ones are $\emptyset$, $\{1,2,3\}$, $\{1,4,5\}$, $\{2,3,4,5\}$, which is G_5. For $k = 6, 7, 8$ the complements of the index sets of Θ_k in $\{1, \dots, k\}$ have rank $k - t_k$: for $k = 6$, $\{4,5,6\}, \{2,3,6\}, \{1,3,5\}$, rank 3; for $k = 7$, $\{4,5,6,7\}, \{2,3,6,7\}, \{1,3,5,7\}, \{2,3,4,5\}$, rank 3; for $k = 8$ the same with 8 added to each, rank 4. So the solutions form a group of 2^{t_k} elements containing G_k, hence equal to G_k, even without the symmetry condition.

Step 2: the flips. In a $(J_1, \dots, J_k)$-space V the involutions of Θ_k split V into joint eigenspaces V_ε, $\varepsilon \in \{\pm 1\}^{t_k}$, and J_α maps V_ε onto $V_{\varepsilon f_\alpha}$, f_α recording which θ's anti-commute with J_α (table of §4). Let F_k be the group generated by $f_1, \dots, f_k$. If V is minimal, the sum of the V_ε over a coset of F_k is invariant, so the non-zero V_ε form exactly one coset. By the table, F_k is everything for $k = 4$ (J_4), 5 (J_2, J_4), 6 (J_1, J_2, J_4) and 8 (J_1, J_2, J_4, J_8); F_3 is trivial; F_7, generated by the flips of J_1, J_2, J_4, is the index-two subgroup $\varepsilon_3\varepsilon_5\varepsilon_7 = 1$. As $\omega_3 = \theta_3$ and $\omega_7 = \pm\theta_3\theta_5\theta_7$ ($\{1,2,3\}\triangle\{1,4,5\}\triangle\{1,6,7\} = \{1, \dots, 7\}$), the two cosets for $k = 3, 7$ are told apart by ω_k, which commutes with every J_α and is therefore $\pm I$ on a minimal space.

Step 3: uniqueness. Let V, V' be minimal with $V_\varepsilon \ne 0 \ne V'_\varepsilon$ for some ε (for $k \not\equiv 3 \pmod 4$ any ε will do, by Step 2; for $k = 3, 7$ this means the same sign of ω_k). Take unit vectors $u \in V_\varepsilon$, $u' \in V'_\varepsilon$. If J_S is skew, $\langle J_S u, u\rangle = 0$. If J_S is symmetric and anti-commutes with some $\theta \in \Theta_k$, then $\langle J_S u, u\rangle = \langle J_S \theta u, \theta u\rangle = \langle \theta J_S \theta u, u\rangle = -\langle J_S u, u\rangle = 0$. Otherwise $S \in G_k$ by Step 1, and $\langle J_S u, u\rangle$ is a universal sign times a product of entries of ε. The same holds for u', and Lemma 5.2 gives $V \cong V'$.

Step 4: dimension. For u as in Step 3, $\langle J_S u, J_T u\rangle = \pm\langle J_{S\triangle T} u, u\rangle$ vanishes unless $S\triangle T \in G_k$, in which case $J_S u = \pm J_T u$. So one $J_S u$ for each coset of G_k in the group of subsets of $\{1, \dots, k\}$ gives an orthonormal basis of V: $\dim V = 2^{k - t_k} = m_k$.

Step 5: existence. $\mathbb{R}^{16}$ with $J'_1, \dots, J'_k$ contains a minimal space (Lemma 5.1). For $k = 3, 7$, replacing J_k by $-J_k$ changes the sign of ω_k, so both classes occur. □

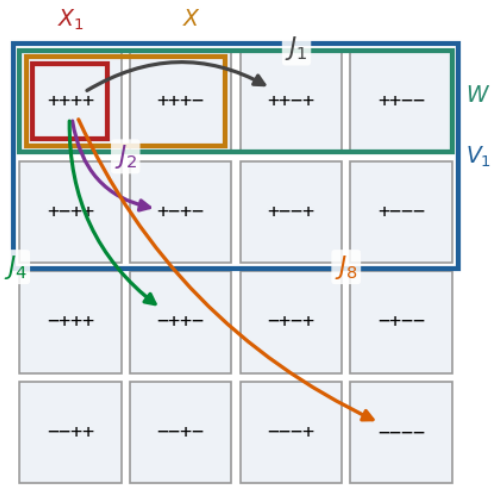
For $k = 8$, Step 4 shows that $\mathbb{R}^{16}$ is itself minimal, with all sixteen joint eigenspaces lines, as used in §4. Milnor's minimal spaces are: lines over $\mathbb{R}, \mathbb{C}, \mathbb{H}$ ($k \le 2$); H, H' with $J_3 = \pm J_1 J_2$ ($k = 3$); $H \oplus H'$ ($k = 4, 5, 6$); L, L' with $J_7 = \pm J_1 \cdots J_6$ ($k = 7$); $L \oplus L'$ ($k = 8$). In the language of Clifford algebras (quoted from Atiyah, Bott and Shapiro, *Clifford modules*, Topology 3 (1964); Lawson and Michelsohn, *Spin Geometry*, Ch. I), a $(J_1, \dots, J_k)$-space is a module over the algebra C_k generated by $e_1, \dots, e_k$ with $e_\alpha^2 = -1$ and $e_\alpha e_\beta = -e_\beta e_\alpha$; only the last two rows (m_k and the number of classes of minimal spaces), proved above, are used ($\mathbb{F}(N)$ denotes $N \times N$ matrices over $\mathbb{F}$):

k	0	1	2	3	4	5	6	7	8
C_k	$\mathbb{R}$	$\mathbb{C}$	$\mathbb{H}$	$\mathbb{H} \oplus \mathbb{H}$	$\mathbb{H}(2)$	$\mathbb{C}(4)$	$\mathbb{R}(8)$	$\mathbb{R}(8) \oplus \mathbb{R}(8)$	$\mathbb{R}(16)$
m_k	1	2	4	4	8	8	8	8	16
classes	1	1	1	2	1	1	1	2	1

An interactive look at the sixteen pieces

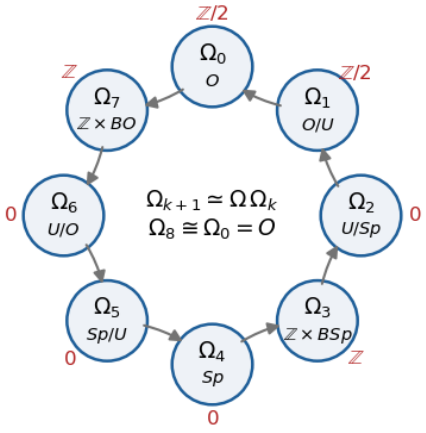
The figure shows $\mathbb{R}^{16}$ with $J_1', \dots, J_8'$ (§3) as a 4×4 grid of sixteen lines, the joint eigenspaces of $\theta_3, \theta_5, \theta_6, \theta_7$, labelled by their signs. Drag the slider k from 0 to 8: only the θ's with indices $\le k$ are defined, and cells on which they agree form one block, giving $1, 1, 1, 2, 2, 4, 8, 16, 16$ blocks, the subspaces V_1, W, X, X_1 of Lemma 24.6. Tap a button J_α ($\alpha \le k$) to see how it permutes the blocks, reversing the signs of the θ's it anti-commutes with. The panel shows $\Omega_k(16r)$, m_k, the number of classes of minimal spaces, and $\pi_k O \cong \pi_0 \Omega_k$. At $k = 8$ the flips of J_1, J_2, J_4, J_8 carry any cell to any other, which is why $\mathbb{R}^{16}$ is minimal.

(In the printed version this figure is a still image; the interactive version is in the web edition.)



signs of $(\theta_3, \theta_5, \theta_6, \theta_7)$ on the 16 lines of $\mathbb{R}^{16}$

J_1, J_2, J_4, J_8 reach every line from X_1: $\mathbb{R}^{16}$ is minimal, $m_8 = 16$



red: $\pi_0\Omega_k \cong \pi_k O$; arrows: one loop each step

6. Proof of Theorem 24.5 (Milnor pp. 144–148)

Fix $1 \le k \le 7$, $n = 16r$, $J = J_k$; the case $k = 0$ is Theorem 24.3, whose range $n - 4$ tends to infinity. By Lemma 24.4 the geodesics of $\Omega_k(n)$ from J are $t \mapsto J\exp(\pi tA)$, $A \in \mathcal{T} = \mathcal{T}_J$, ending at $-J$ iff $\exp(\pi A) = -I$; the component of J is a symmetric space whose curvature, being totally geodesic, is that of $O(n)$ (Lemma 2.2(b)). Translating Theorem 21.3(b) to J by the isometry L_J, $R(JA, JB)JA = J \cdot \frac{1}{4}[[A, B], A]$; so for the velocity $V = \pi JA$, $K_V(JB) = \pi^2 J\, K_A(B)$ with

$$K_A(B) = -\tfrac{1}{4}[A, [A, B]] = \tfrac{1}{4}(-A^2B + 2ABA - BA^2), \qquad B \in \mathcal{T},$$

and by Theorem 20.5 an eigenvalue e^2 of K_A gives conjugate points at $t = s/e$, $s = 1, 2, \ldots$. **If $e^2 \ge 4$ there is one at $t = 1/e \le 1/2$**, so the index is at least the number of eigenvalues ≥ 4, counted with multiplicity.

Lemma 6.1 (splitting). Let $A \in \mathcal{T}$ with $\exp(\pi A) = -I$. Then $\mathbb{R}^n = M_1 \oplus \cdots \oplus M_s$ orthogonally, each M_h invariant and minimal under $\mathcal{F} = \{J_1, \ldots, J_{k-1}, J, A\}$, and $A^2 = -a_h^2$ on M_h with a_h a positive odd integer. The geodesic is minimal iff all $a_h = 1$.

Proof. All members of $\mathcal{F}$ are skew, so the splitting exists as in Lemma 5.1. A^2 is symmetric, negative semi-definite and commutes with $\mathcal{F}$, so its eigenspaces in M_h are invariant and $A^2|M_h = -a_h^2$ (Milnor's "all equal, except for sign"). If $a_h = 0$, then $\exp(\pi A)|M_h = I$; otherwise A/a_h is a complex structure on M_h and $\exp(\pi A)|M_h = \cos(\pi a_h)I + \sin(\pi a_h)A/a_h = -I$ iff a_h is odd. Minimality means $A^2 = -I$ (Lemma 24.4(iii)). □

Case $k \not\equiv 2 \pmod 4$

Assertion. The index of a non-minimal geodesic from J to $-J$ in $\Omega_k(n)$ is $\ge n/m_{k+1} - 1$.

Proof. On M_h put $J_h' = a_h^{-1}JA$, a complex structure anti-commuting with $J_1, \ldots, J_{k-1}$ and J: $(JA)^2 = J(AJ)A = -J^2A^2 = -a_h^2$, $(JA)^{\mathsf{T}} = AJ = -JA$, $J_\alpha JA = -JAJ_\alpha$, $J(JA) = -A = -(JA)J$. Since $A = -a_h JJ_h'$ on M_h, M_h is minimal for the $k+1$ structures $J_1, \ldots, J_{k-1}, J, J_h'$, so $\dim M_h = m_{k+1}$ and $s = n/m_{k+1}$. For $h \ne j$, $(J_1, \ldots, J_{k-1}, -J, J_j')$ also makes M_j minimal; as $k + 1 \not\equiv 3 \pmod 4$ and $k + 1 \le 8$, Lemma 24.8 gives an isometry $C : M_h \to M_j$ (Milnor's "isomorphism from M_h to $\bar{M}_j$") with

$$CJ_\alpha = J_\alpha C \ (\alpha < k), \qquad CJ = -JC, \qquad CJ_h' = J_j'C.$$

Let $B = C$ on M_h, $-C^{\mathsf{T}}$ on M_j, 0 on the other M_l; C^{T} satisfies the same relations from M_j to M_h, so $B \in \mathcal{T}$. For $v \in M_h$, $BAv = -a_h BJJ_h'v = a_h JJ_j'Bv = -(a_h/a_j)ABv$ (as $A = -a_j JJ_j'$ on M_j), so $ABAv = a_h a_j Bv$ and

$$(K_AB)v = \tfrac{1}{4}(a_j^2 + 2a_h a_j + a_h^2)Bv = \tfrac{1}{4}(a_h + a_j)^2\, Bv,$$

and likewise on M_j. If the geodesic is not minimal, some $a_{h_0} \ge 3$, and the $s-1$ eigenvectors from the pairs (h_0, j) are independent, with eigenvalues $(a_{h_0} + a_j)^2/4 \ge 4$. So the index is $\ge s - 1$. □

Hence, by Theorem 22.1 with $\Omega^d \cong \Omega_{k+1}(n)$ (Lemma 24.4), φ_r induces isomorphisms of π_i for $i \le n/m_{k+1} - 3$, a range that tends to infinity with r.

Case $k \equiv 2 \pmod 4$ ($k = 2, 6$)

The difficulty (Milnor): $\Omega\Omega_k(n)$ has infinitely many components, $\Omega_{k+1}(n)$ finitely many. Let $i = J_1 \cdots J_{k-1}$; as $k - 1 \equiv 1 \pmod 4$, Lemma 2.1 makes i a complex structure commuting with $J_1, \dots, J_{k-1}$ and anti-commuting with every $J' \in \Omega_k(n)$. Make $\mathbb{R}^n$ complex by i. For $J' \in \Omega_k(n)$, $J^{-1}J'$ is unitary; put $f(J') = \det_{\mathbb{C}}(J^{-1}J') \in S^1$. Each $A \in \mathcal{T}$ commutes with i and is skew, hence skew-Hermitian, and $f(J \exp(\pi t A)) = e^{\pi t \operatorname{tr}_{\mathbb{C}} A}$. Also $f(-J) = (-1)^{n/2} = 1$, as $n/2 = 8r$ is even; Milnor's "closed loop" needs this.

The components. For a path ω from J to $-J$ let ϑ_ω be the continuous argument of $f \circ \omega$ with $\vartheta_\omega(0) = 0$, so $\vartheta_\omega(1) \in 2\pi\mathbb{Z}$. If ω' is uniformly close to ω, then $|f\circ\omega - f\circ\omega'| < 2$, so $|\vartheta_\omega - \vartheta_{\omega'}| < \pi$ throughout and $\vartheta_\omega(1) = \vartheta_{\omega'}(1)$. Hence for each integer c the set $\Omega^{(c)}$ of paths with $\vartheta_\omega(1) = \pi c m_k/2$ is open and closed.

The geodesics. Split $\mathbb{R}^n = M_1 \oplus \cdots \oplus M_\nu$ as in Lemma 6.1. The complex eigenspaces of A are invariant under J (if $Av = \lambda i v$, then $A(Jv) = -JAv = \lambda i Jv$) and the J_α, so $A|M_h = a_h i$, a_h an odd integer of either sign. Each M_h is then a minimal $(J_1, \dots, J_{k-1}, J)$-space, $\dim M_h = m_k$, $\nu = n/m_k$ (Milnor's r), and $\operatorname{tr}_{\mathbb{C}} A = i(a_1 + \cdots + a_\nu)m_k/2$: the geodesic lies in $\Omega^{(c)}$ with $c = a_1 + \cdots + a_\nu = -\operatorname{tr}_{\mathbb{R}}(iA)/m_k$ (as $\operatorname{tr}_{\mathbb{R}} = 2\operatorname{Re}\operatorname{tr}_{\mathbb{C}}$). It is minimal iff all $a_h = \pm 1$; such geodesics exist in $\Omega^{(c)}$ iff $|c| \le \nu$ and $c \equiv \nu \pmod 2$, and they form $\varphi(\Omega_{k+1}(n)^{(c)})$, where $\Omega_{k+1}(n)^{(c)} = \varphi^{-1}(\Omega^{(c)})$ is open and closed in $\Omega_{k+1}(n)$. (For "if": split $\mathbb{R}^n$ into minimal $(J_1, \dots, J_{k-1}, J)$-spaces M_h by Lemma 5.1. A minimal $(J_1, \dots, J_{k+1})$-space has dimension $m_{k+1} = m_k$, so it is also minimal for its first k structures, hence isomorphic to each M_h by Lemma 24.8; transporting its last structure gives a complex structure J_h'' on M_h anti-commuting with $J_1, \dots, J_{k-1}, J$. Then $J' = \bigoplus_h \pm J_h''$ lies in $\Omega_{k+1}(n)$, $A = J^{-1}J'$ has $a_h = \pm 1$, and changing the sign on M_h changes the sign of a_h, so every such c occurs.)

Assertion. A non-minimal geodesic in $\Omega^{(c)}$ has index $\lambda \ge (\nu - |c|)/4$.

Proof. For $h \ne j$, Lemma 24.8 for k structures ($k \not\equiv 3 \pmod 4$), applied to M_h and to M_j with J replaced by $-J$, gives an isometry $C : M_h \to M_j$ commuting with $J_1, \dots, J_{k-1}$, hence with i, and anti-commuting with J. With $B = C$ on M_h, $-C^{\mathsf{T}}$ on M_j, 0 elsewhere, $B \in \mathcal{T}$, and for $v \in M_h$: $ABAv = a_h a_j i^2 Bv = -a_h a_j Bv$, so $(K_A B)v = \frac{1}{4}(a_j^2 - 2a_h a_j + a_h^2)Bv$: B is an eigenvector with eigenvalue $(a_h - a_j)^2/4$. For $a_h > a_j$ this gives $(a_h - a_j)/2 - 1$ conjugate points in $(0, 1)$, and the B's of different pairs are independent, so

$$2\lambda \ge \sum_{a_h > a_j} (a_h - a_j - 2).$$

Let p be the sum of the positive a_h and $-q$ that of the negative ones: $p - q = c$ and $p + q \ge \nu$. Non-minimal means some $|a_{h_0}| \ge 3$. If $a_{h_0} \le -3$ (Milnor: "suppose for example that some a_h is equal to -3"), pairing h_0 with the positive a_h gives $2\lambda \ge \sum_{a_h > 0}(a_h + 3 - 2) \ge p \ge (\nu + c)/2$. If $a_{h_0} \ge 3$, pairing with the negative ones gives $2\lambda \ge q \ge (\nu - c)/2$. In either case $4\lambda \ge \nu - |c|$. □

Theorem 22.1 is now needed one component at a time.

Lemma 6.2 (Theorem 22.1 componentwise). Let $\Omega' \subseteq \Omega(M; p, q)$ be open and closed and contain a minimal geodesic. If $\Omega' \cap \Omega^d$ is a topological manifold and every non-minimal geodesic in Ω' has index $\ge \lambda_0$, then $\pi_i(\Omega', \Omega' \cap \Omega^d) = 0$ for $0 \le i < \lambda_0$.

Proof. Milnor's proof of 22.1 (pp. 122–123) applies Lemma 22.5 to $F \circ E$ on the manifold $B = \operatorname{Int}\Omega^c(t_0, \dots, t_k)$, a deformation retract of $\operatorname{Int}\Omega^c$. Intersect with Ω': $B \cap \Omega'$ is open and closed in B; a deformation keeps points in their path components, so preserves Ω'; and $F \circ E$ on $B \cap \Omega'$ attains its minimum exactly on $\Omega' \cap \Omega^d \neq \emptyset$. The proof then runs verbatim. □

Apply Lemma 6.2 to $\Omega' = \Omega^{(c)}$, for fixed c and n so large that $|c| \leq \nu$ (only even c occur, as ν is even): φ maps $\Omega_{k+1}(n)^{(c)}$ into $\Omega^{(c)}$ inducing isomorphisms of π_i for $i \leq (\nu - |c|)/4 - 2$. Under $\omega \mapsto \omega \oplus \delta_k$ the integer c does not change: f is multiplicative on block sums, so c is additive, and δ_k has $c = -\operatorname{tr}(i\, J_k'^{-1} J'_{k+1})/m_k = \operatorname{tr}(\omega_{k+1})/m_k = 0$ on $\mathbb{R}^{16}$, because $\omega_3 = \theta_3$ and $\omega_7 = \pm\theta_3\theta_5\theta_7$ have trace 0 (all sixteen sign patterns occur once). This is Milnor's conclusion: each component of $\Omega\Omega_k(n)$ is approximated, in a range of dimensions that grows with n, by the corresponding component of $\Omega_{k+1}(n)$.

Passing to the limit

In both cases, for each $i \geq 1$ and each base point, φ_r induces isomorphisms on π_i once r is large. On π_0: in the first case φ_r is a bijection once $n/m_{k+1} \geq 3$. In the second, each set $\Omega' = \{\omega : \vartheta_\omega(1) = \text{const}\}$ is open and closed, and if non-empty it contains a geodesic: for $\omega \in \Omega'$ and $E(\omega) < a$, the deformation of Theorem 16.2 moves ω, inside Ω' and without increasing E, into the compact set $B^a \cap \Omega'$, where E attains its minimum at a local minimum, hence a critical point, of $E|_B$. So every path lies in some $\Omega^{(c)}$ with $c \equiv \nu \pmod 2$. The integer c does not change under stabilization, and once $\nu \geq |c| + 8$, Lemma 6.2 with $\lambda_0 \geq 2$ shows that φ_r maps the components of $\Omega_{k+1}(16r)^{(c)}$ bijectively onto those of $\Omega^{(c)}$. As φ_{r+1} extends φ_r and homotopy groups and path components of the colimits are colimits (§3), Theorem 24.5 follows.

> **Milnor's text quotes two facts here.** To get a homotopy equivalence Milnor asserts, without proof, that both Ω_{k+1} and $\Omega\Omega_k$ "have the homotopy type of a CW-complex", and applies Whitehead's theorem. We quote the same (colimits of compact manifolds along smooth embeddings are homotopy equivalent to mapping telescopes; loop spaces by J. Milnor, Trans. AMS 90 (1959)). Every homotopy-group statement of this chapter, including $\pi_i O \cong \pi_{i+8} O$, is proved without them.

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7. Application: vector fields on spheres and real vector bundles over spheres

Vector fields (Milnor's Remark, pp. 143–144). If $J_1, \dots, J_k$ are anti-commuting complex structures on $\mathbb{R}^n$ and $|x| = 1$, then $J_1 x, \dots, J_k x$ are orthonormal and orthogonal to x: $\langle J_\alpha x, x\rangle = 0$ since J_α is skew, and $\langle J_\alpha x, J_\beta x\rangle = -\langle J_\beta J_\alpha x, x\rangle = 0$ since $J_\beta J_\alpha$ is skew. So S^{n-1} carries k orthonormal tangent fields. By Lemmas 5.1 and 24.8, $\mathbb{R}^n$ carries k such structures iff $m_k \mid n$; with $m_{k+8} = 16 m_k$, the largest such k for $n = 2^{4a+b} \cdot (\text{odd})$ is $8a + 2^b - 1 = \rho(n) - 1$. Milnor's examples: 3 fields on S^{4r-1}, 7 on S^{8r-1}, 8 on S^{16r-1}; concretely, $x \mapsto J'_1 x, \dots, J'_8 x$ (§3) on S^{15}. For $n = 2, 4, 8, 16, 32, 64, 128, 256$,

$$\rho(n) - 1 = 1,\ 3,\ 7,\ 8,\ 9,\ 11,\ 15,\ 16,$$

growing by 8 each time n is multiplied by 16. Adams (*Vector fields on spheres*, Annals of Math. 75 (1962)) proved that there are never $\rho(n)$ independent fields; his proof uses real K-theory, whose foundation is Theorem 24.7. So S^{n-1} is parallelizable only for $n = 1, 2, 4, 8$; and for n odd Adams's bound is the hairy ball theorem.

Real vector bundles over spheres. As in Chapter 14, §5, real vector bundles over S^n are glued from two trivial ones by maps $S^{n-1} \to O(N)$, and the stable classes form the group $\widetilde{KO}(S^n) \cong \pi_{n-1}O$. By Theorem 24.7 and the table of §4:

n (mod 8)	1	2	3	4	5	6	7	0
$\widetilde{KO}(S^n)$	$\mathbb{Z}/2$	$\mathbb{Z}/2$	0	$\mathbb{Z}$	0	0	0	$\mathbb{Z}$

For $n = 1, 2, 4, 8$ a generator is glued by $x \mapsto x_0 I + \sum_{\alpha<n} x_\alpha J_\alpha$, built from a minimal space for $n-1$ structures, of dimension $m_0, m_1, m_3, m_7 = 1, 2, 4, 8$ (orthogonal for $|x| = 1$, by anti-commutation): the Möbius band and the real bundles underlying the complex, quaternionic and octonionic Hopf line bundles (Atiyah, Bott and Shapiro, 1964). So every real vector bundle over S^3, S^5, S^6, S^7 becomes trivial after adding a trivial bundle, while over S^{8k+4} and S^{8k} there is an integer invariant and over S^{8k+1} and S^{8k+2} a $\mathbb{Z}/2$ invariant (w_1 and w_2 for $k = 0$).

The eightfold way of free fermions. In Kitaev's periodic table of topological insulators and superconductors (AIP Conf. Proc. 1134 (2009)), the phases in the eight real symmetry classes are classified by π_0 of Milnor's $\Omega_0, \dots, \Omega_7$, that is, of $O, O/U, U/Sp, \mathbb{Z} \times BSp, Sp, Sp/U, U/O, \mathbb{Z} \times BO$, shifted along the chain by the dimension; Kitaev's Majorana wire of 2001 carries the $\mathbb{Z}/2 = \pi_0 O$. The period of the table is Theorem 24.7.

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8. Summary of what was silently assumed, now made explicit

Where	Milnor says	What is actually needed
p. 133	$O(n)$ "inherits" a bi-invariant metric; $\Omega_1(n) \cong O(n)/U(n/2)$	Chapter 14's chart argument for real matrices; transitivity from the normal form
pp. 133–134	normal form of skew matrices; "homeomorphic to the space $\Omega_1(n)$"	real spectral theorem (quoted); $\gamma_J = \cos \pi t\, I + \sin \pi t\, J$, compactness
p. 135	"Computation shows" the eigenvalues of K_A	the splitting $M_2(\mathbb{R}) = C_+ \oplus C_-$ of 2×2 blocks (Lemma 24.2)
p. 136	Lemma 24.4 "for $0 \le l < k$"; components are symmetric spaces	$\Omega_{l+1}(n) \neq \emptyset$, $J_0 := I$; $I_J(J') = -JJ'J$ preserves $\Omega_k(n)$
p. 137	Assertions 1, 2 "iff"; "this clearly implies … totally geodesic"	"only if" needs A small; Lemma 2.2(a), from Theorem 10.4 and Corollary 10.7
pp. 137–138	minimal in $O(n)$ "certainly minimal in $\Omega_k(n)$"; the converse	distances in $\Omega_k(n)$ dominate those in $O(n)$; the assertions applied for small t
p. 138	limit inclusions "not difficult to see"	the block $\mathbb{R}^{16}$ with explicit $J'_1, \dots, J'_8$; $\varphi_{r+1}(J' \oplus J'_{k+1}) = \varphi_r(J') \oplus \delta_k$
p. 138	Theorem 24.5 "for each $k \ge 0$"	proved here for $k \le 7$ (Milnor treats Lemma 24.8 only for ≤ 8 structures, p. 143)
pp. 139–142	Lemma 24.6: "clear", "not hard to see", "it can be shown"	Lemmas 4.1, 4.2 (odd and even steps), Lemma 4.3 (the pieces)

Where	Milnor says	What is actually needed
pp. 139, 141	"restrict attention to the component of largest dimension"	J_3, J_7 must be balanced, otherwise Ω_4, Ω_8 are empty
p. 142	the table "left to the reader"	$\pi_k O \cong \pi_0 \Omega_k$; group structures from $\pi_1(U/Sp)$ and $\pi_3 Sp(1)$
p. 143	Lemma 24.8, "follows that of 24.6"; $m_k = 16m_{k-8}$ "can be shown"	proof by joint eigenvectors and cyclic vectors (§5); the footnote's identification (§4)
p. 144	splitting into minimal M_h; K_A computed "just as before"	complements of invariant subspaces are invariant; Gauss equation (Lemma 2.2(b))
p. 145	"Such an isomorphism exists by 24.8"	the structures with $-J$ are again minimal; $k+1 \leq 8$
p. 146	both spaces "have the homotopy type of a CW-complex"; $\Omega_k(n)$ "has an infinite cyclic fundamental group"	quoted, needed only for homotopy types; the second is not needed (the integer c)
p. 147	f maps the geodesic "into a closed loop"	$f(-J) = (-1)^{n/2}$, so $n/2$ must be even
p. 148	"restrict attention to some fixed component"; "for example … some a_h is equal to -3"	$\Omega^{(c)}$ open and closed, Theorem 22.1 componentwise (Lemma 6.2), c stable; $\lvert a_h \rvert \geq 3$ of either sign gives $4\lambda \geq \nu - \lvert c \rvert$

Checked by machine. Milnor's §24 has been verified in the Lean proof assistant, for each n and, for the periodicity theorem, in the stable range. The algebra, for real matrices and abstract vector spaces: the compactness of $\Omega_k(n)$; the algebraic parts of Lemma 24.4 (ii) and (iii); Assertions 1 and 2, with their converses for small A; the odd steps of Lemma 24.6 as bijections onto sets of involutions (Lemma 4.1), and the dimensions of Lemma 4.3; explicit anti-commuting complex structures on $\mathbb{R}^4$, $\mathbb{R}^8$ and $\mathbb{R}^{16r}$, with Milnor's embedding $\Omega_k(n) \to \Omega_k(n+n')$; Lemma 24.2 and the index bounds of §6 from the spectrum of K_A; Lemma 24.8 for $k \leq 8$ (minimal spaces have dimension m_k and, for $k \not\equiv 3 \pmod 4$, are unique up to a linear isomorphism commuting with the J_α); and the vector fields on spheres of §7. The geometry and the topology: each stratum of $\Omega_k(n)$, the part where $\mathcal{T}_J$ has a given dimension, is a compact submanifold of $M_n(\mathbb{R})$ whose geodesics are the curves $t \mapsto J\exp(tA)$, $A \in \mathcal{T}_J$ (Lemma 24.4 (i); the geodesics are found directly, without Lemma 2.2); Lemma 24.1 and Lemma 24.4 (iii), with the distance $\pi\sqrt{n}$ obtained from the Hopf–Rinow theorem in the stratum, and with $\Omega_{k+1}(n)$ homotopy equivalent to the space of minimal paths (homeomorphic to it after identifying paths that agree on $[0,1]$); an explicit neighbourhood retraction onto the minimal paths, so that Hanner's theorem is not needed; the splitting of Lemma 6.1; the bounds of Lemma 24.2 and of both Assertions of §6 for the index of the geodesics themselves, from Jacobi fields given by isometries of $\Omega_k(n)$ and the Index Theorem, without the curvature of $\Omega_k(n)$; Theorem 24.3; Lemma 6.2, and that every path from J to $-J$ lies in exactly one $\Omega^{(c)}$, c an integer; Theorem 24.5 for each n and $k \leq 7$, as bijections of homotopy groups in a range that grows with n (for $k = 2, 6$ on each $\Omega^{(c)}$, with the path components for $\lvert c \rvert \leq \nu - 8$); the compatibility $\varphi_{r+1}(J' \oplus J'_{k+1}) = \varphi_r(J') \oplus \delta_k$ and the invariance of c; Lemma 24.6-(8), $\Omega_8(16r) \cong O(r)$; and Theorem 24.7 in the form $\pi_i O(n) \cong \pi_{i+8} O(n')$ for $n \geq i+2$ and $n' \geq i+10$, a bijection of sets. Not checked: the colimits Ω_k and O and the passage to the limit; the homotopy equivalences, which rest on the two quoted facts; the group structure along the chain; the table of §4 and the identifica-

tions of Lemma 24.6-(3) to (7) with Grassmannians and classical groups; uniqueness up to isometry and the two classes for $k = 3, 7$ in Lemma 24.8; the eigenvalues of K_A on $\Omega_k(n)$ (Lemma 2.2 (b)); the symmetric-space structure of Lemma 24.4 (ii); and the rest of §7. See the verified edition.